

Deep Assessment Report

Residential Real Estate SPV Notes — Luxembourg / Germany

A case-specific assessment report for tokenization viability and value creation.

September 22, 2026

A TVAC assessment by Astoval

CLARITY
BEFORE
COMMITMENT.

About this report

TVAC evaluates whether tokenizing a specific asset / structure creates net added value, using the five-factor model:

$$\text{Added Value} = (\text{New Opportunities (NO)} + \text{Cost Savings (CS)} + \text{Risk Reduction (RR)}) - (\text{Tokenization Costs (TC)} + \text{New Risks (NR)})$$

Where this report fits

TVAC is designed for the very early decision stage of a tokenization initiative. It tests whether the proposed use of tokenization appears likely to create genuine added value before significant resources are committed to legal structuring, issuance-platform or provider selection, detailed technical architecture, or implementation. A positive verdict is therefore not implementation approval; it indicates that the case may justify deeper legal, commercial, technical and operational work.

Case description summary

The initiating organization is a privately owned real estate holding company owning three stabilized residential assets in Berlin and Hamburg through German GmbHs. The proposed case is intended to raise EUR 15–20 million to finance renovations, partially refinance bank debt, and strengthen liquidity buffers. Instead of raising additional traditional mezzanine capital, the organization is evaluating tokenization through a Luxembourg SPV issuing profit-participating notes linked to net rental cash flow after operating costs and senior debt service. We plan a 12-month lock-up followed by structured quarterly liquidity windows on a regulated venue. The properties generate steady rental income but require energy upgrades, façade improvements, and selected tenant-area refurbishments to remain competitive and compliant with evolving ESG standards.

The full user-supplied case description and structured input record are reproduced in Appendix E.

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Methodology	TVAC — Tokenization Value-Add Calculator
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01 | Executive summary

TOKENIZATION VALUE VERDICT

Conditional Go

ADDED VALUE

-3.00

plus 8.00 minus 11.00

Headline: This case is sufficiently defined for a meaningful case-specific assessment, but it is not yet a finalized execution design. The assessment lands at Conditional Go with an estimated Net Added Value of -3.00 (Plus: 8.00 - Minus: 11.00).

Important interpretation note: This verdict concerns tokenization added value only. A Viable or Conditional Go verdict does not mean that the underlying asset is a good investment, that projected returns are realistic, that credit risk is acceptable, or that the instrument is suitable for any investor.

Case design stage: Structured Concept

This assessment is based on a partially defined design. The case is sufficiently clear for a meaningful case-specific evaluation, but several structural choices remain open.

1.1 What is different about this case

The initiating organization is a privately owned real estate holding company owning three stabilized residential assets in Berlin and Hamburg through German GmbHs. The Luxembourg SPV wrapper may widen access beyond a purely domestic financing route by targeting professional investors in both Germany and the Netherlands rather than only the company's existing lender set. This structure appears to require several additional workstreams for the Asset Owner at once: Luxembourg issuance architecture, cross-border distribution design, custody, venue coordination and rights documentation.

KEY TAKEAWAYS

- 01 Key findings:** The Luxembourg SPV wrapper may widen access beyond a purely domestic financing route by targeting professional investors in both Germany and the Netherlands rather than only the company's existing lender set. Across New Opportunities, Cost Savings and Risk Reduction, the combined Plus score is 8.00.
- 02 Key constraints:** This structure appears to require several additional workstreams for the Asset Owner at once: Luxembourg issuance architecture, cross-border distribution design, custody, venue coordination and rights documentation. Across Tokenization Costs and New Risks, the combined Minus score is 11.00.
- 03 Decision posture:** Proceed only if the remaining material design choices are finalized before build-out. The case is not yet ready to move directly into implementation; it still requires disciplined design completion around investor scope, compliance controls, custody/settlement, and operating model.
- 04 Impact:** If those remaining design choices are finalized well, the case has a plausible path to a stronger value profile by reducing New Risks (NR)/Tokenization Costs (TC) and making the identified benefits more likely to survive implementation. If they are not, the current value case remains too fragile for straightforward execution.

02 | Asset class & market context

The initiating organization is a privately owned real estate holding company owning three stabilized residential assets in Berlin and Hamburg through German GmbHs. The proposed case is intended to raise EUR 15–20 million to finance renovations, partially refinance bank debt, and strengthen liquidity buffers. This is a rights, ownership-structure, distribution and servicing case. Tokenization should not be treated as creating liquidity by itself; the relevant test is whether the proposed structure improves investor access, administration, governance or transaction processes relative to a conventional SPV, fund or direct-ownership arrangement.

Current market phase: Case-derived structural context only; external market maturity is not assessed from verified market sources for this case.

What this context means for the case

Against this market backdrop, the points below show which features of the proposed structure support the value case and which still constrain it.

Structural factors supporting the case:

- The Luxembourg SPV wrapper may widen access beyond a purely domestic financing route by targeting professional investors in both Germany and the Netherlands rather than only the company's existing lender set.
- A regulated custodian and whitelist-based onboarding may reduce some transfer-control and holder-identification risk compared with a looser private-transfer model.
- The case describes a legal or economic wrapper that can be assessed for rights mapping, governance and lifecycle administration.
- A controlled investor scope can support a more bounded initial design than unrestricted public distribution.

Structural factors that still constrain the case:

- This structure appears to require several additional workstreams for the Asset Owner at once: Luxembourg issuance architecture, cross-border distribution design, custody, venue coordination and rights documentation.
- The case adds cross-border legal, tax and operational dependency risk because cash flows originate from German residential assets while the instrument is issued through Luxembourg and sold into Germany and the Netherlands.
- Property rights, issuer or SPV rights, authoritative registers and investor entitlements must remain aligned across legal documentation and the token layer.
- Secondary demand, valuation, transfer friction and the full legal and administrative cost stack must be compared with realistic conventional alternatives.
- The multi-jurisdiction sale scope increases the need to separate common design logic from local distribution, investor-category, marketing and transfer requirements.

No live market data or unverified model memory was used for this market-context section. Where no verified market source is activated, current adoption, prevailing practice and market phase remain unassessed.

03 | Analysis result

ADDED VALUE

-3.00

PLUS

8.00

MINUS

11.00

Verdict rationale:

This case lands at Conditional Go because the current structure produces an Added Value of -3.00: the strongest supportive driver is New Opportunities (5/10), while the main friction is Tokenization Costs (7/10). The Luxembourg SPV wrapper may widen access beyond a purely domestic financing route by targeting professional investors in both Germany and the Netherlands rather than only the company's existing lender set. This structure appears to require several additional workstreams for the Asset Owner at once: Luxembourg issuance architecture, cross-border distribution design, custody, venue coordination and rights documentation. This rationale concerns tokenization added value only and does not validate legal, regulatory, tax, financial, investment, credit, yield, issuer, or implementation suitability.

Factor score profile

Factor	Score	Case-specific driver
New Opportunities	5/10	The Luxembourg SPV wrapper may widen access beyond a purely domestic financing route by targeting professional investors in both Germany and the Netherlands rather than only the company's existing lender set.
Cost Savings	0/10	Long-term administrative efficiency is stated as an objective, but no validated baseline compares the tokenized route with a conventional mezzanine or private-note process for the same financing need.
Risk Reduction	3/10	A regulated custodian and whitelist-based onboarding may reduce some transfer-control and holder-identification risk compared with a looser private-transfer model.
Tokenization Costs	7/10	This structure appears to require several additional workstreams for the Asset Owner at once: Luxembourg issuance architecture, cross-border distribution design, custody, venue coordination and rights documentation.
New Risks	4/10	The case adds cross-border legal, tax and operational dependency risk because cash flows originate from German residential assets while the instrument is issued through Luxembourg and sold into Germany and the Netherlands.

Added Value: -3.00 (plus 8.00 / minus 11.00)

What still needs to be verified:

These are the main points that should be independently confirmed before moving from concept toward implementation.

- No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
- Obtain written jurisdiction-specific legal analysis of classification, offering pathway, rights mapping and transfer controls.
- Document and validate the enforceable link between the token, underlying rights, ownership records and cash-flow waterfall.

Regulatory status: Luxembourg · Case-specific applicability unverified

TVAC has identified relevant regulatory context for the stated issuance jurisdiction and case-specific legal questions, but it has not verified how the law applies to the final structure. Final legal treatment should therefore be confirmed by qualified local counsel before implementation.

Sources used and outstanding validation questions are summarized in Appendix A.5.

The score becomes useful only when the structure behind it is coherent. The sections below test the value pathways, participant or investor perimeter, operating burden, regulatory path, custody and transfer fit, and—where material—the settlement and cash-leg design that must support the claimed benefits.

3.1 Value pathways

This section isolates the mechanisms through which tokenization could create incremental value over a realistic conventional alternative. It is not an assessment of whether the underlying asset or financing need is attractive in its own right.

- The core business purpose is capital formation for three stabilized German residential assets, not property-title digitization: the proposed token would finance energy upgrades, façade works, tenant-area refurbishments, partial bank-debt refinancing and liquidity buffers.
- The main upside thesis is narrower and more credible than a full retail-liquidity story: the Luxembourg SPV note could package contractual participation in net rental cash flow and principal repayment for a cross-border professional-investor perimeter in Germany and the Netherlands.
- Tokenization may support better investor transparency through a more structured rights layer, whitelisted holder base and cleaner reporting around rental cash flow, debt service and renovation use of proceeds.
- Fractional access is stated as an objective, but in this case it appears aimed at smaller professional tickets rather than broad retail distribution; that limits reach, but also keeps the investor base closer to institutional real-estate credit buyers.

- The proposed quarterly liquidity windows are potentially helpful, but they remain a design intention rather than demonstrated liquidity; without documented venue access and investor participation, this remains an opportunity hypothesis rather than a realized value driver.

3.2 Investor segmentation

Investor scope affects both upside and friction. A narrower professional or institutional audience can reduce distribution and compliance complexity, but it can also limit the access or liquidity benefit that tokenization is expected to create.

- The investor strategy is clearly non-retail. That reduces some distribution complexity compared with a public mass-market offer, but it also means the product must stand up against other professional real-estate credit or structured income options.
- The most coherent target segment appears to be professional investors seeking contractual exposure to German residential cash flow with no property-level operating control and with some potential for smaller ticket sizes than a conventional bespoke mezzanine placement.
- Because the note is profit-participating and linked to residual cash flow after operating costs and senior debt service, investors will likely focus on downside sequencing, renovation reporting and rate sensitivity rather than token format alone.

- The additional 'accredited' label remains unresolved in local legal mapping and should not drive the design until qualified local advice confirms how the intended investor perimeter is described in each sale jurisdiction.
- No evidence is provided on anchor investors, minimum ticket strategy, or whether the Germany/Netherlands investor base specifically prefers a tokenized wrapper over a conventional private note.

3.3 Operational costs

This section distinguishes plausible process savings from the additional cost of introducing the token layer. A saving becomes a strong value driver only when it can be compared with the current process and is not offset by data, integration, provider or compliance work.

- The structure introduces likely cost categories at several layers: Luxembourg SPV work, transaction documentation, onboarding and KYC/AML, regulated custody, issuance and registry tooling, venue coordination, reporting and cross-border tax analysis.
- Those costs arise before the Asset Owner captures any demonstrated efficiency benefit, which is the main reason the case is constrained despite a sensible professional-investor design.
- The user states long-term administrative efficiency as an objective, but no validated baseline has been provided showing that this tokenized route is cheaper than a conventional Luxembourg or German private-placement process for the same EUR 15–20 million financing goal.
- Property-level reporting requirements do not disappear in a tokenized structure. Renovation tracking, rental-income reporting, debt-service visibility and investor communications still need to be maintained with institutional discipline.
- Because the properties are already owned through German GmbHs, the token layer is an additional financing architecture rather than a replacement for an existing fragmented ownership process.

3.4 Regulatory path

This is regulatory scoping, not a legal opinion. The purpose is to identify the verified regulatory context and the design questions that materially affect the case, while making clear what still requires qualified jurisdiction-specific validation.

- TVAC has approved baseline regulatory context for Luxembourg, but it has not activated a verified case-specific local claim. The frameworks below are used

for structured scoping only; applicability and current detailed conditions remain to be independently confirmed.

- EU scoping should first distinguish a token that is a financial instrument or other regulated financial product from a crypto-asset governed by MiCA. MiCA does not replace the EU financial-services framework for instruments already covered by that framework, and tokenization does not itself determine classification.
- Where the token represents a transferable security or another financial instrument, MiFID II, the Prospectus Regulation and related national rules may shape distribution, investor access, intermediary roles and disclosure. The final facts determine which requirements or exemptions are relevant.
- The DLT Pilot Regime provides an EU framework for specified DLT market infrastructures, while MiFID II, CSDR and national implementation remain relevant to trading, settlement, custody, recordkeeping and market-infrastructure functions according to the final design.
- EU-level analysis must be combined with member-state law and supervisory practice for issuer or SPV law, property and security rights, authoritative registers, insolvency, tax, marketing and operational roles. An EU framework reference does not resolve those national questions.
- For real-estate tokenization, the financial-instrument and distribution analysis must be separated from the national legal treatment of the property, the issuer or SPV, investor rights and land or ownership records. The token layer should not be assumed to replace legally authoritative national records.
- Approved baseline context identifies the following potentially relevant frameworks or regimes for scoping in Luxembourg: EU securities and markets framework (MiFID II / Prospectus / etc.) (EU baseline); EU crypto-asset framework (MiCA); EU fund/management frameworks (as relevant); AML regime (Luxembourg / EU); Data protection (GDPR). Their case-specific applicability, current conditions and interaction must be independently confirmed.
- Issuer structure and investor base often determine the dominant regulatory path.
- Distribution/marketing constraints can be a primary design driver for cross-border setups.
- Custody/safeguarding and governance controls can be decisive for institutional participation.
- Germany: approved baseline scoping context is available for distribution, investor eligibility, marketing, transfer and intermediary-role analysis. It

does not determine local applicability or compliance.

- Germany — baseline scoping: Approved baseline context identifies the following potentially relevant frameworks or regimes for scoping in Germany: EU securities and markets framework (MiFID II / Prospectus / etc.) (EU baseline); German electronic securities framework (general); AML regime (Germany / EU); Data protection (GDPR). Their case-specific applicability, current conditions and interaction must be independently confirmed.
- Germany — baseline scoping: EU baseline applies; local supervisory practice and implementation details can matter.
- Germany — baseline scoping: Tokenized instruments that resemble securities typically inherit securities offering/distribution constraints.
- Netherlands: approved baseline scoping context is available for distribution, investor eligibility, marketing, transfer and intermediary-role analysis. It does not determine local applicability or compliance.
- Netherlands — baseline scoping: Approved baseline context identifies the following potentially relevant frameworks or regimes for scoping in Netherlands: EU securities and markets framework (MiFID II / Prospectus / etc.) (EU baseline); EU crypto-asset framework (MiCA); AML regime (Netherlands / EU); Data protection (GDPR). Their case-specific applicability, current conditions and interaction must be independently confirmed.
- Netherlands — baseline scoping: Classification and distribution perimeter remain the main gating factors.
- Netherlands — baseline scoping: Marketing constraints and investor communication discipline matter for compliance outcomes.

3.5 Custody & venue fit

Custody and venue design should be proportionate to the value thesis. If the case does not depend on continuous secondary trading, extra custody, venue or transfer complexity should be justified by a concrete operational, control or distribution benefit.

- A regulated custodian is a logical fit for the stated professional-investor perimeter and can support stronger control expectations than a self-custody model for this type of real-estate-linked note.
- The planned quarterly liquidity windows are more realistic than continuous-liquidity claims for a stabilized residential cash-flow instrument, but they still require an operating model that links custodian, venue, transfer restrictions and payment processes.

- The user-selected 'MTF or other regulated venue (user-selected interface label; local classification unconfirmed)' label should be treated cautiously in EU-facing analysis. A multilateral trading facility or other regulated-venue pathway may be relevant, but the final venue mapping remains unconfirmed.
- T+1/2 settlement is proportionate to the case's value proposition because the case claims transparency, access and administrative efficiency more than real-time settlement benefits.
- Custody and venue fit are directionally sensible, but not yet anchored by named counterparties, binding scope or tested workflows.

3.6 Settlement & Cash-Leg Readiness

Settlement is assessed against the case's actual value logic, not against an assumption that tokenization requires digital money or T+0. The key question is whether the cash leg can operate reliably for the intended investors and operating model without unnecessary complexity.

- Why it matters: A material part of the claimed value depends on the settlement design working reliably for the intended participants.
- Current position: the stated payment / settlement model is T+1/T+2 securities settlement. Its readiness is Defined conventional design. This is a design/readiness assessment, not evidence that settlement finality has already been achieved.
- What must be proven: asset- and cash-leg finality, synchronization, failed-settlement handling, exception management and recovery. Until those points are evidenced, settlement-dependent benefits should remain conditional rather than treated as realized outcomes.
- Assessment basis: this treatment reflects the stated asset/instrument type (wrapper) and liquidity intent (quarterly); TVAC does not assume that every tokenized asset requires advanced digital money or T+0 settlement.

04 | Stakeholder value & alignment

This chapter asks whether the parties needed to make the structure work are economically and operationally aligned. It focuses on the value each party receives, the burden and responsibility it carries, and the dependencies that could weaken the project's overall Added Value.

4.1 Value & alignment overview

This view does not re-score the five TVAC factors for each party. It shows instead what each stakeholder stands to gain, what burden or responsibility it carries, and what must be aligned for the project value case to hold.

Asset owner / Issuer (sponsor, manager)

Expected value depends mainly on whether a Luxembourg-anchored income-linked structure marketed into Germany, Netherlands can improve funding flexibility and investor proposition without cross-border compliance, periodic-liquidity design, and venue/custody costs overwhelming Asset Owner economics.

Value received

The main upside comes from alternative capital access, a clearer income-linked investor proposition and more disciplined investor administration than a realistic conventional non-tokenized alternative.

Cost, risk & responsibility

Higher upfront legal/structuring work to ensure enforceability of token-holder rights through the wrapper.

Alignment test

Lock the operating model: rights (term sheet), custody setup, venue path, cash-flow/distribution mechanics, and transfer rules.

Investors

Expected value depends mainly on whether investors can gain a clearer income-linked exposure through a Luxembourg note setup marketed across Germany, Netherlands, with better reporting and a user-stated transfer process that is independently validated rather than assumed to provide broad market liquidity.

Value received

The intended investor-side value is access to an income-linked real-asset exposure with potential improvements in reporting and transfer discipline; any advantage over a conventional non-tokenized alternative remains to be demonstrated.

Cost, risk & responsibility

Need extra clarity on legal enforceability: what exactly the token represents and how claims are exercised.

Alignment test

Demand a clean transfer-process note covering the operator role, the explicitly selected quarterly cadence, transfer restrictions, settlement mechanics and what happens if secondary demand is shallow.

Infrastructure providers

Expected value depends on whether providers can support a Luxembourg-anchored wrapper distributed across Germany, Netherlands, with reporting, custody and any user-stated transfer mechanics that are operationally credible without assuming secondary-market depth.

Value received

Infrastructure providers may benefit if they can support a legally and operationally validated wrapper, reporting, custody, and user-defined transfer stack for cross-border income-linked instruments.

Cost, risk & responsibility

Integration and operational complexity (reconciliations, reporting, corporate actions, incident response) often scales faster than expected.

Alignment test

Define interfaces and responsibilities early (RACI): who owns KYC/AML, whitelists, transfers, reconciliations, and reporting.

4.2 Asset owner / Issuer (sponsor, manager)

Value received

- The main upside comes from alternative capital access, a clearer income-linked investor proposition and more disciplined investor administration than a realistic conventional non-tokenized alternative.
- The user-selected transfer environment may support a more structured transfer process, but no legal characterization, market depth or liquidity outcome is assumed.
- The selected custody model may support investor confidence and onboarding discipline if its legal and operational role is validated.
- If tokenization demonstrably unlocks capital, financing flexibility or terms that are not realistically available through bank debt, a club deal or a conventional private placement, the initiator's economics can change materially after successful capital formation.

Costs, risks & responsibilities

- Higher upfront legal/structuring work to ensure enforceability of token-holder rights through the wrapper.
- Cross-border compliance complexity across Luxembourg (issuance) + Germany, Netherlands (sales) (marketing rules, investor eligibility, reporting).
- Listing/admission, ongoing disclosure expectations, and vendor/venue dependency can increase OPEX.
- Whitelist controls can constrain liquidity and require continuous compliance operations.

Alignment & actions

- Lock the operating model: rights (term sheet), custody setup, venue path, cash-flow/distribution mechanics, and transfer rules.
- Resolve the highest-priority evidence gaps identified in the report before spending further on build or integration.
- Stress-test the explicitly selected quarterly transfer process: define the operator role, approvals, settlement steps and what happens if matching demand is shallow. No liquidity support is assumed.
- Define a financing-evidence plan: compare target amount, timing, investor access, financing terms and full costs against the realistic conventional alternative before treating tokenization value as proven.

4.3 Investors

Value received

- The intended investor-side value is access to an income-linked real-asset exposure with potential improvements in reporting and transfer discipline; any advantage over a conventional non-tokenized alternative remains to be demonstrated.
- The selected regulated-custodian design may support safeguarding and auditability if account, ownership and control arrangements are validated.
- A clearer secondary-trading route than typical private market holdings (but liquidity depth is not guaranteed).

Costs, risks & responsibilities

- Need extra clarity on legal enforceability: what exactly the token represents and how claims are exercised.
- Transfer restrictions (whitelists) can limit who you can sell to, and when.
- Cross-border compliance can reduce distribution breadth and may introduce withholding/reporting complexity.
- Information asymmetry risk if disclosure is light—investors will demand compensating safeguards (audits, reporting, governance).

Alignment & actions

- Demand a clean transfer-process note covering the operator role, the explicitly selected quarterly cadence, transfer restrictions, settlement mechanics and what happens if secondary demand is shallow.
- Verify custody and operational controls in operational detail: account model, beneficial ownership evidence, transfer-approval workflow, incident handling, auditability, and how investor rights are evidenced across the custody, registry and operating stack.
- Clarify economics: fee stack (custody/venue/platform), expected reporting cadence, and how cash flows are distributed.

4.4 Infrastructure providers

Value received

- Infrastructure providers may benefit if they can support a legally and operationally validated wrapper, reporting, custody, and user-defined transfer stack for cross-border income-linked instruments.
- The selected custody model may be relevant to the stated investor scope, but demand and legal necessity are not assumed.
- Venue participation can expand revenue sources (listing, surveillance, member onboarding, reporting).

Costs, risks & responsibilities

- Integration and operational complexity (reconciliations, reporting, corporate actions, incident response) often scales faster than expected.
- Regulatory perimeter and liability exposure can increase with cross-border sales, investor-scope expansion, and more complex distribution structures.
- Whitelist logic can interfere with matching/settlement and requires coordination across venue + custodian + issuer.

Alignment & actions

- Define interfaces and responsibilities early (RACI): who owns KYC/AML, whitelists, transfers, reconciliations, and reporting.
- Agree evidence expectations: SOC reports, audit trails, code audits, incident runbooks, and SLA/exit clauses.
- Align commercial terms with actual delivery responsibilities and evidenced outcomes; avoid adding a token layer without measurable operational or commercial value.

4.5 Who bears the costs and risks?

In a capital-formation case, the issuer may carry substantial structuring, legal, compliance and distribution costs before funding is secured. That is a pre-funding economic position, not necessarily the final outcome. The relevant question is whether tokenization demonstrably produces incremental capital access, financing flexibility or terms that outweigh the added stack relative to a realistic conventional alternative.

Case-specific conclusion: Under the current initial configuration, and before capital formation, distribution commitments and cost allocation are proven, this tokenization appears commercially more attractive to service providers than to the initiating Asset Owner.

- A materially negative pre-funding issuer position should trigger a demand, terms and cost-allocation review, not an automatic conclusion that tokenization can never work. It becomes a warning sign if investor demand remains uncommitted, the same capital could reasonably be raised conventionally on similar terms, or provider fees absorb the incremental financing benefit.
- If investor net position is low or negative, expect weak demand unless enforceability, disclosures, custody, and realistic exit paths are improved.

05 | Factor breakdown – detailed explanation

This chapter explains the causal basis of the five scores: which facts, assumptions and evidence gaps create upside, and which ones introduce additional friction. The three positive factors ask what tokenization adds relative to a realistic conventional alternative; the two negative factors ask what extra cost or risk the token layer introduces. A low positive score therefore means limited or weakly evidenced tokenization benefit, while a low negative score means limited additional friction. The scores are calibrated methodology judgments, not measured outcomes or legal conclusions.

Added Value = (New Opportunities + Cost Savings + Risk Reduction) – (Tokenization Costs + New Risks)

Overall, the case is finely balanced, with modeled upside and modeled friction close to each other in magnitude. The structure is directionally viable, but specific design conditions and mitigations need to be met before committing to a full-scale rollout. With an Added Value score of -3, the case remains condition-sensitive and should be judged by whether the key gating conditions can be locked in. Net Added Value is primarily driven by new opportunities at 5/10 on the upside, and constrained mainly by tokenization costs at 7/10 on the friction side.

New opportunities (NO)

- The Luxembourg SPV wrapper may widen access beyond a purely domestic financing route by targeting professional investors in both Germany and the Netherlands rather than only the company's existing lender set.
- Whitelist-based transfer controls, regulated custody and stated contractual economic rights create a more structured investor product than an informal bilateral mezzanine arrangement, especially for smaller professional tickets.
- The opportunity case is still capped because the planned quarterly liquidity windows and venue access are not yet backed by documented operating commitments or demonstrated investor demand.
- The token does not convey operational control over the Berlin and Hamburg properties, so the upside is primarily distribution, transparency and process design rather than a fundamentally new real-estate right.

Cost savings (CS)

- Long-term administrative efficiency is stated as an objective, but no validated baseline compares the tokenized route with a conventional mezzanine or private-note process for the same financing need.
- For this EUR 15–20 million raise, new layers such as Luxembourg SPV work, custody, venue coordination and cross-border tax analysis may absorb any near-term workflow savings.
- Digital reporting and registry processes may help over time, but that benefit has not been demonstrated in this specific Berlin and Hamburg portfolio financing case.
- Because the assets are already held through German GmbHs, tokenization adds a financing layer rather than replacing a clearly inefficient legacy investor-administration stack.

Risk reduction (RR)

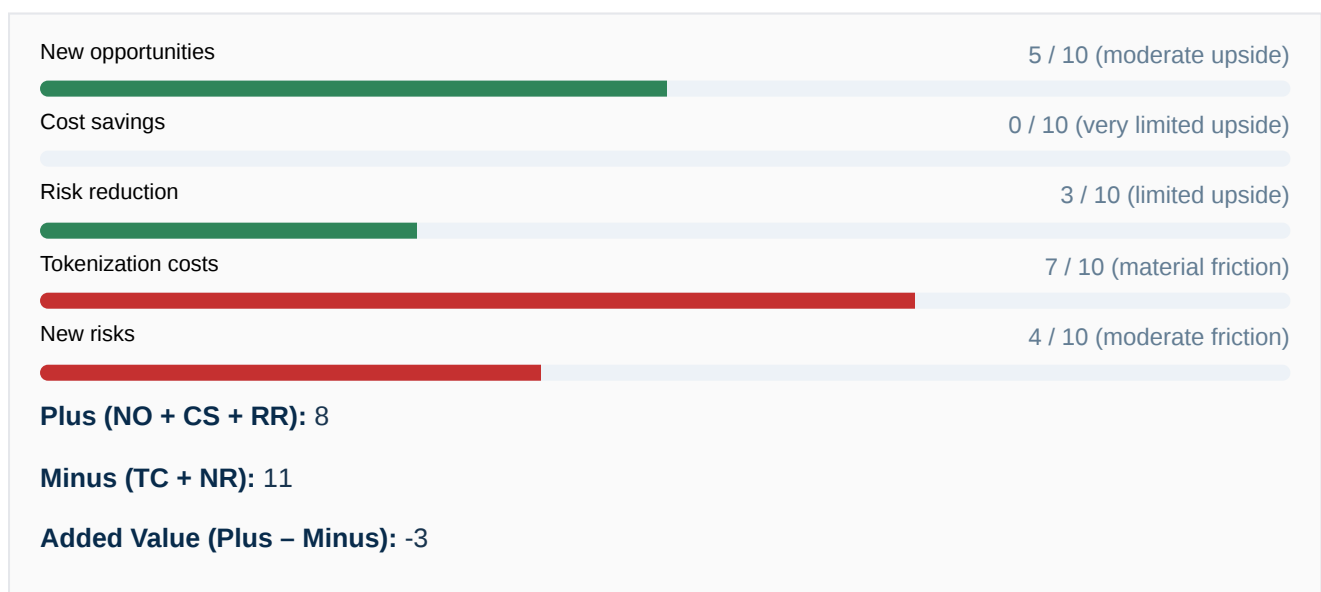
- A regulated custodian and whitelist-based onboarding may reduce some transfer-control and holder-identification risk compared with a looser private-transfer model.
- The stated contractual rights are narrower and cleaner than offering token holders operational control over the underlying properties, which helps keep governance boundaries clearer.
- Investor transparency may improve if rental cash flow, senior debt service and renovation use of proceeds are reported in a disciplined manner through the tokenized note structure.
- Risk reduction remains limited because tax, venue, rights mapping and renovation execution still introduce meaningful uncertainty that tokenization does not remove.

Tokenization costs (TC)

- This structure appears to require several additional workstreams for the Asset Owner at once: Luxembourg issuance architecture, cross-border distribution design, custody, venue coordination and rights documentation.
- The professional-investor scope reduces some retail complexity, but it does not remove the likely burden of multi-jurisdiction analysis across Luxembourg, Germany and the Netherlands.
- Quarterly liquidity windows on a regulated venue add operating and provider complexity beyond a static private placement.
- At the stated financing size, those added layers look heavy relative to the still-unproven efficiency benefit.

New risks (NR)

- The case adds cross-border legal, tax and operational dependency risk because cash flows originate from German residential assets while the instrument is issued through Luxembourg and sold into Germany and the Netherlands.
- The liquidity story could disappoint if the planned venue path and investor participation do not support meaningful secondary activity after the 12-month lock-up.
- The note depends on accurate mapping between token terms, legal documentation, custodian records and payment processes; that legal-technical interface is not yet evidenced.
- New risks are material but not obviously fatal because the design is permissioned, non-retail, and aligned with conventional settlement timing rather than a high-friction 24/7 DeFi model.



What could change this verdict

These are the highest-impact conditions that could materially strengthen or weaken the current assessment. They are practical decision gates, not predictions.

- Distribution scope and investor segmentation (jurisdiction count, retail vs professional posture)
- Rights mapping and structural clarity (asset/SPV structure, cash-flow logic, investor protections)
- Custody / controls / transfer restrictions (institutional-grade requirements vs complexity)
- Liquidity assumptions (timing, venue feasibility, realistic market depth)
- Payment / cash-leg readiness where the case depends on DvP, digital cash, 24/7 settlement, collateral mobility or automated lifecycle payments
- Tokenization cost profile (legal/compliance/ops overhead and non-scalable components)

06 | Key risks & mitigation

This matrix prioritizes the report's identified risks by impact and likelihood. It is not a measured probability model; each position should be read together with the case-specific description and mitigation.

	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 High			R2 R5		
4 Med-high			R1 R3 R4 R6		
3 Medium					
2 Med-low					
1 Low					

LIKELIHOOD →

Risk zones are indicative prioritization bands, not measured probabilities.

PRIORITY RISKS

R1 Misaligned incentives / vendor-capture risk

Impact 4 · Likelihood 3

Mitigation: Use stage-gating and milestones (legal opinions + demand validation) before committing; negotiate fee structures with meaningful success-based components; ensure portability/exit clauses if the platform/venue changes strategy.

R2 Cross-border tax leakage or withholding mismatch

Impact 5 · Likelihood 3

Mitigation: Obtain independent Luxembourg, German and Dutch tax analysis for issuer cash flows, investor payments and withholding treatment before launch

R3 Renovation underperformance reduces distributable cash flow

Impact 4 · Likelihood 3

Mitigation: Document capex governance, reporting triggers and downside cases linking renovation progress to investor disclosure

R4 Quarterly liquidity windows prove weaker than expected

Impact 4 · Likelihood 3

Mitigation: Validate venue workflow, transfer mechanics and investor demand before relying on liquidity in marketing or structuring

R5 Rights mapping and authoritative records remain unclear

Impact 5 · Likelihood 3

Mitigation: Obtain written mapping of note terms, token logic, custodian records and any issuer or registrar records

R6 Interest-rate sensitivity weakens valuations or investor appetite

Impact 4 · Likelihood 3

Mitigation: Stress-test cash-flow coverage and investor disclosures under adverse financing and valuation scenarios

07 | Recommendations & next steps

Case-specific focus: This recommendation set assumes a Luxembourg-led issuance structure aimed at professional and accredited/qualified investors. Primary execution friction: The final classification, rights mapping, offering structure, investor eligibility, marketing perimeter, ownership record, transfer controls and servicing responsibilities require qualified jurisdiction-specific validation. The recommendations below therefore focus on reducing avoidable friction before execution.

7.1 Design priorities and decision gates

Design stage: Structured Concept. Recommendations below focus on the highest-impact unresolved design and evidence gates.

The Conditional Go reflects a genuinely balanced, design-dependent case: the positive factors (NO 5, CS 0, RR 3) sit close to Tokenization Costs and New Risks (TC 7, NR 4) and must be read together rather than as a simple positive signal. For the initiating Asset Owner, value capture is still constrained by high upfront and ongoing structuring / operating costs, non-trivial execution and market-adoption risk, and a distribution design that relies too heavily on assumptions rather than locked-in commitments. In this case, a cross-border setup (2 sale jurisdictions (Germany, Netherlands)) and distribution requirements are likely key drivers of both TC and NR. The recommendations below therefore focus on the few design changes and evidence gaps most likely to determine whether tokenization becomes economically stronger or loses its marginal advantage.

Primary clarification and refinement priorities

The priorities below are selected from the largest score drag and the unresolved evidence or operating-model gates in this case.

7.1.1 Treat Distribution and Liquidity as a Design Requirement, Not an Assumption

because the liquidity pathway is currently uncommitted and increases execution risk.

In this case specifically: The distribution and liquidity path needs to be credible in practice, not just conceptually attractive.

Compared with a conventional alternative: Compared with a conventional distribution model, the tokenized structure must improve actual investor access, transferability, or control — not just add technical complexity.

- Secure anchor investors and/or indicative investor commitments before finalising the issuance structure and vendor stack.
- Require the selected distribution or transfer providers to document the responsibilities, eligibility checks, approval steps, settlement process and reporting duties that are explicitly in scope.

- Do not assume a transfer cadence, matching mechanism, indicative quoting, liquidity support or exit route that the user has not specified. Where the input is silent, mark the mechanism unresolved.

Expected effect: Reduced New Risks (NR) related to liquidity shortfall and improved credibility of the investment case for both Asset Owner and investors.

7.1.2 Rebalance Cost and Risk Allocation Away from the Asset Owner

because Tokenization Costs are currently a dominant downside driver.

In this case specifically: The proposed issuance is anchored in Luxembourg with distribution into Germany, Netherlands; the Asset Owner therefore needs explicit control of cross-jurisdiction cost, responsibility and execution burden.

Compared with a conventional alternative: Compared with a conventional implementation path, tokenization only earns its place here if the added stack does not push too much fixed cost and execution burden back onto the Asset Owner.

- Reduce fixed upfront fees and shift a meaningful portion of platform, custody and venue remuneration to success-linked components (e.g., issuance completion, minimum capital raised, or documented post-issuance delivery milestones explicitly stated in the case).
- Narrow the initial scope to a single issuance phase, deferring optional features until value is demonstrated.
- Explicitly cap ongoing post-issuance service costs (custody, venue, reporting, corporate actions) to avoid open-ended operational drag.

Expected effect: Lower Tokenization Costs (TC) for the Asset Owner and improved net Added Value without changing the core structure.

7.1.3 What a successful redesign would need to change

The measures above target the factors that currently keep this case at Conditional Go. They do not by themselves imply a future Viable verdict. A stronger result would require evidence that Tokenization Costs and New Risks have actually fallen, while the identified benefits remain credible against the conventional alternative. That should be tested in a new assessment once the relevant design choices and evidence have changed.

In this case specifically: The decisive question is whether the structure can move from a plausible concept to a tightly controlled operating model across Germany, Netherlands.

7.2 Design what-if scenarios

These illustrative scenarios explore how near-term design choices – for example narrowing the set of sale jurisdictions, adjusting the investor mix, or changing custody and venue setup – could move the Added Value score up or down for this specific case. They are not predictions about the broader future of the asset class, but levers you can pull in the current structuring and implementation of this transaction.

- If the initial investor perimeter is tightened to clearly defined professional investors only, the distribution story may become easier to evidence and the compliance and communications burden may narrow.
- If the quarterly liquidity windows are deferred from the initial launch and treated as a later feature, tokenization costs and operating complexity may fall, but part of the access and tradability thesis would weaken.
- If the structure is used primarily for renovation financing and investor reporting clarity rather than also emphasizing refinancing and liquidity-buffer flexibility, the rights narrative may become easier to communicate and underwrite.

- If the rights-and-records mapping is documented early and kept economically narrow, the product may be easier to distribute than a broader token design with additional governance features.

7.3 Suggested next steps

- **No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.**

A before-and-after baseline is needed to show whether tokenization actually improves the current process rather than merely changing the technology stack.

- **Obtain written jurisdiction-specific legal analysis of classification, offering pathway, rights mapping and transfer controls.**

The assessment can identify the legal question, but implementation should not proceed until the relevant rights, roles and legal effect are confirmed for the final structure.

- **Document and validate the enforceable link between the token, underlying rights, ownership records and cash-flow waterfall.**

The assessment can identify the legal question, but implementation should not proceed until the relevant rights, roles and legal effect are confirmed for the final structure.

- **Which offering/distribution posture is intended in each sale jurisdiction, and is the operating model aligned with that posture?**

This turns an open assumption into evidence that can support the next decision gate.

- **Which regulated roles (onboarding, custody, venue, agent/registrar) are expected or preferred for the chosen design?**

Clear ownership prevents gaps between the token layer and existing legal, custody, settlement and operational processes.

Appendices

- Appendix A — Method & interpretation
- Appendix B — Operational roadmap
- Appendix C — TVAC Provider Directory (illustrative)
- Appendix D — Glossary & abbreviations
- Appendix E — User-supplied case inputs
- Appendix F — Disclaimer

Appendix A — Method & interpretation

A.1 How to use this appendix

This appendix explains how to interpret the assessment, what evidence is still needed, and which conditions could strengthen or weaken the case. It is not a technical manual for TVAC.

A.2 What the assessment establishes — and what it does not

This report provides informational decision support only. It does not constitute legal, regulatory, tax, financial, or investment advice, and it does not provide a compliance determination, valuation, or provider endorsement.

Case snapshot (as stated)

- Issuance jurisdiction: Luxembourg
- Sale jurisdictions / target markets: Germany, Netherlands
- Asset / instrument type (indicative): wrapper
- Liquidity intent (as stated): quarterly
- Payment / cash-leg indication: T+1/T+2 securities settlement
- Cash-leg dependency / readiness: High dependency / Defined conventional design readiness

What the Tokenization Value Verdict means

- Viable: Tokenization added value is plausible under the stated assumptions, subject to normal verification and execution.
- Conditional Go: Tokenization added value is design-dependent. Proceeding without meeting the listed conditions materially increases execution risk.
- Non-viable: Under current assumptions, tokenization is unlikely to outperform a conventional structure on a risk-adjusted basis.

What TVAC did not assess

- No definitive legal, regulatory, tax, or investment conclusions
- No asset valuation, yield projections, or market-price forecasts
- No assurance of secondary-market liquidity or investor demand
- No operational readiness audit beyond the indicative checklist

A.3 Minimum evidence before the next major commitment

Before proceeding toward implementation, decision-makers should be able to substantiate a minimum level of evidence across the areas below. This checklist is not exhaustive and does not imply regulatory sufficiency. It reflects common governance expectations in institutional-grade projects.

Highest-priority gaps in this case

(derived from inputs/completeness)

1. No verified case-specific regulatory claim is activated; approved baseline context supports structured scoping, while case-specific legal application remains unresolved.
2. Obtain written jurisdiction-specific legal analysis of classification, offering pathway, rights mapping and transfer controls.
3. Document and validate the enforceable link between the token, underlying rights, ownership records and cash-flow waterfall.

Structural & economic foundations

- Clear description of the underlying asset and ownership structure
- Defined economic rights and cash-flow logic
- Draft term sheet or structural outline
- Clear rationale for tokenization vs alternatives

Regulatory & distribution posture

- Issuance jurisdiction clearly defined
- Explicit list of sale jurisdictions
- Documented investor type assumptions
- Initial regulatory scoping (high-level)

Token & control design

- High-level token design and rights mapping
- Transfer restriction and whitelist logic (conceptual)
- Corporate actions and investor reporting approach
- Governance for smart-contract or protocol changes (if relevant)

Infrastructure & counterparties

- Custody model (conceptual)
- Indicative partner shortlist (non-binding)
- Issuance, registry, and lifecycle approach
- Payment / cash-leg and settlement approach, proportionate to the case's actual value proposition
- Early assessment of secondary trading (if any)

Cost & operational realism

- High-level CAPEX and OPEX view
- Identification of major cost drivers
- Clear internal ownership of ongoing operations
- Understanding of scalable vs non-scalable costs Lack of clarity across several of these areas materially increases execution risk, even where the conceptual case appears attractive.

A.4 What could strengthen or weaken the case

- Upside driver: New opportunities (value depends on distribution and product/instrument design)
- Risk driver: New risks (most sensitive to regulatory interpretation, custody/ops complexity, and cross-border distribution)
- Cost driver: Tokenization costs (most sensitive to legal/compliance scope, integrations, and ongoing ops overhead)

High-impact decision levers (practical)

- Distribution scope: reduce jurisdictions and/or tighten investor segments to lower compliance and operational overhead.
- Liquidity posture: treat early secondary trading as optional; defer until structure and reporting are stable.
- Custody & controls: choose a control model that meets institutional requirements without over-engineering.
- Payment / cash-leg: clarify the settlement model only to the level required by the claimed value proposition; do not over-engineer it for low-dependency cases.
- Rights & governance: tighten rights mapping (cash flows, reporting, governance, transfer rules) to reduce friction and disputes.

How to use this

If you need the verdict to improve, change the few levers above rather than adding features. In many cases, viability increases by reducing ambition and complexity.

A.5 Evidence & source transparency

Basis of assessment

The assessment is based on the customer's case description and structured inputs, the fixed TVAC methodology, and any activated verified external sources. User-supplied operational and commercial claims have not been independently verified unless explicitly stated.

External sources used

- Directive 2014/65/EU on markets in financial instruments · European Union · checked 2026-07-15

- Regulation (EU) 2017/1129 — Prospectus Regulation · European Union · checked 2026-07-15
- Regulation (EU) 2023/1114 — Markets in Crypto-assets · European Union · checked 2026-07-15
- Regulation (EU) 2022/858 — DLT Pilot Regime · European Union · checked 2026-07-15
- Regulation (EU) No 909/2014 — Central Securities Depositories Regulation · European Union · checked 2026-07-15

What remains unverified

- Which offering/distribution posture is intended in each sale jurisdiction, and is the operating model aligned with that posture?
- Which regulated roles (onboarding, custody, venue, agent/registrar) are expected or preferred for the chosen design?
- If secondary trading is planned, what is the feasible venue pathway for the target markets?

Appendix B — Operational roadmap

This roadmap translates the assessment into a staged implementation path for the specific case type. It is not a project plan or implementation instruction: each phase should be adapted to the final legal structure, counterparties, providers and evidence available. Progress only when the decision gate for the preceding phase has been met.

B.1 Operational roadmap (structured)

The items below are indicative planning scaffolding. They are not a substitute for jurisdiction-specific legal, tax, financial, and operational due diligence, and they must be tailored to the selected asset type, wrapper structure, investor segment, custody model, venue, and settlement design.

B.2 Phase 1 — Preparation & feasibility

Case-specific focus for this phase: Primary issuance jurisdiction: Luxembourg. Initial distribution scope: Germany, Netherlands. Initial investor focus: professional investors, other eligible non-retail investors (local classification unconfirmed).

Objectives

- Clarify the underlying asset/instrument, economic rights, and proposed token wrapper.
- Define target investor segments (professional / eligible / retail, if applicable) and distribution scope.
- Establish internal alignment on objectives, success criteria, and risk appetite.

Key decisions

- Token rights model (equity-like, debt-like, fund-like, revenue share, participation note, etc.).
- Transfer restriction regime (whitelisting, holding periods, jurisdictional gating).
- Custody and settlement direction (account-based vs. wallet-based; T+0 vs. T+1+).

Deliverables (checklist)

- One-page case definition: asset, wrapper, jurisdiction(s), investor segment, and target size.
- Preliminary term sheet outline (rights, restrictions, fees, lifecycle events, disclosures).
- High-level implementation estimate (time bands, cost bands, key dependencies).
- Initial risk register (top 10–15 risks with owners and first mitigations).

Primary stakeholders

- Issuer/asset owner (business sponsor, finance, legal/compliance, operations/IT).
- External legal counsel (securities, funds, structuring), tax adviser, and (if relevant) auditor.
- Shortlisted tokenization platform/technology provider, regulated custodian, and potential venue(s).

Dependencies / gates

- Clear legal characterization of the instrument and its regulatory perimeter in the issuance jurisdiction.
- Agreement on must-have constraints (investor eligibility, reporting, custody, settlement, data handling).

- Internal approval to proceed to structured design (budget and governance gate).

Risk hotspots

- Unclear instrument classification or incompatible distribution intent (e.g., retail scope without a verified pathway).
- Mismatch between user-stated transfer expectations and the legally and operationally supportable process.
- Hidden operational burdens (cap-table/admin, reporting, corporate actions, tax documentation).

B.3 Phase 2 — Legal, tax & regulatory structuring

Case-specific focus for this phase: Core legal and tax analysis should start from Luxembourg. Distribution permissions must be validated for Germany, Netherlands. Generic TVAC disclosure label selected in the input: Exempt. The applicable local legal category remains to be confirmed. Transfer-control design to validate: Whitelist + EU gating + 12-month lock-up.

Objectives

- Obtain independent legal validation of a legally supportable issuance and distribution structure for the intended jurisdiction(s).
- Define disclosure, governance, and ongoing compliance obligations for the lifecycle of the instrument.
- Obtain qualified tax analysis of the proposed wrapper and cash-flow mechanics for issuer and investors.

Key decisions

- Issuance vehicle choice (direct issuer vs. SPV; fund wrapper vs. corporate wrapper).
- Offering and documentation pathway verified for the stated jurisdiction(s); do not infer a legal category from interface labels.
- Investor onboarding model (KYC/AML responsibilities, reliance model, delegated vs. in-house).
- Choice of legal documentation stack verified by qualified local advisers.

Deliverables (checklist)

- Legal characterization memo and jurisdiction matrix (requirements by country/investor segment).
- Draft offering documentation and investor disclosures.
- Transfer restriction policy and rulebook (whitelist logic, jurisdictional gating, and any explicitly selected holding-period rules).
- Tax memo(s) covering withholding, reporting, and cross-border distribution considerations.
- Compliance operating model (roles, escalation paths, record-keeping, audit trail).

Primary stakeholders

- Issuer legal/compliance, external counsel, tax advisers.
- Custodian/compliance function (if regulated custody is used), transfer agent (if applicable).
- Any transfer provider or environment explicitly retained in scope and independently validated.

Dependencies / gates

- Independent legal confirmation of the offering pathway and distribution permissions across the stated target jurisdictions.
- Operational feasibility of investor onboarding and transfer controls (including data privacy constraints).
- Formal sign-off on documentation pack and compliance model.

Risk hotspots

- Jurisdiction conflicts (marketing rules, suitability requirements, local registration triggers).
- Under-scoped ongoing obligations (periodic reporting, investor communications, corporate actions).
- Tax frictions that erode the added value thesis (withholding, reporting, complexity costs).

B.4 Phase 3 — Build, integration & vendor onboarding

Case-specific focus for this phase: The current custody design indicates regulated custodian. Its precise legal and operational role remains to be confirmed. The current transfer environment indicates ATS or MTF; this does not establish a local legal venue classification. The current settlement design indicates T+1/T+2 securities settlement. Its operational and legal fit remains to be validated. Transfer-rule implementation must reflect: Whitelist + EU gating + 12-month lock-up.

Objectives

- Finalize the technology stack and integrate custody, settlement, onboarding, and reporting workflows.
- Implement transfer rules and lifecycle event handling (issuance, redemptions, distributions, corporate actions).
- Establish operational readiness (controls, monitoring, incident response, reconciliation).

Key decisions

- Token standard / ledger choice and compatibility with custody and venue requirements.
- Settlement model (atomic delivery-versus-payment vs. coordinated settlement through intermediaries).
- Data and reporting architecture (who stores what; audit trail; privacy boundaries).
- Outsourcing vs. in-house responsibility split (transfer agent, admin, reporting).

Deliverables (checklist)

- Final technical architecture and integration plan (custody, venue, payments, identity/KYC).
- Rule engine / whitelist implementation and test suite (including jurisdictional gates).
- Operational runbooks (reconciliation, incident response, corporate actions, investor support).
- UAT results and sign-offs (issuer, custodian, venue, admin/TA as relevant).
- Security review (key management model, access control, change management, monitoring).

Primary stakeholders

- Issuer operations/IT, tokenization platform/tech provider, regulated custodian where explicitly selected.
- Selected transfer-environment integration team, payment rails (if applicable), admin/TA providers.
- Compliance and risk functions for control design and approval.

Dependencies / gates

- Signed vendor contracts, SLAs, and responsibility matrix (RACI) across the stack.
- Completed integration testing and reconciliation proof (end-to-end lifecycle flows).
- Readiness gate: controls, monitoring, and support operations in place.

Risk hotspots

- Integration complexity and reconciliation gaps across custody/venue/admin systems.
- Rule enforcement drift (transfer restrictions not consistently applied across venues/wallets).
- Operational single points of failure (provider concentration, key management, support load).

B.5 Phase 4 — Issuance, distribution & launch

Case-specific focus for this phase: Any legally required launch permissions and onboarding constraints must be independently confirmed for: Germany, Netherlands. User-selected generic investor segment for launch planning: professional investors, other eligible non-retail investors (local classification unconfirmed). Local legal category mapping remains to be confirmed. The launch transfer environment currently indicates ATS or MTF; this does not establish a local legal venue classification. The current liquidity intent is quarterly transfer windows. No market status, matching demand or liquidity support is inferred.

Objectives

- Execute issuance mechanics and investor onboarding under the independently verified offering pathway.
- Operationalize disclosures, subscriptions, allocations, and post-issuance record-keeping.
- Operationalize only the user-stated and legally validated transfer process, if any; no venue listing, market making or liquidity support is assumed.

Key decisions

- Distribution channel mix (direct, intermediated, platform marketplace, placement agents) as legally validated.
- Subscription mechanics and settlement timing (funding rails, DVP, cutoff times) as actually specified.
- Transfer model, if any: stated approvals, eligibility, timing and limitations; no cadence or support mechanism is assumed.

Deliverables (checklist)

- Final investor communication pack and distribution checklist.
- Subscription and allocation runbook; reconciliation and exception handling.
- Go-live monitoring plan and incident escalation paths.
- Documented transfer-process setup, if applicable, and evidence-based monitoring metrics.

Primary stakeholders

- Issuer distribution team, platform operator, onboarding/KYC provider.
- Custodian, settlement and transfer providers explicitly selected for the stated process.
- Legal/compliance for marketing controls and ongoing disclosure obligations.

Dependencies / gates

- Approved marketing/distribution materials and compliance sign-off.
- End-to-end dry run of subscription, allocation, and post-issuance reporting.
- Operational readiness gate: support, monitoring, and governance in place.

Risk hotspots

- Distribution frictions (slow onboarding, suitability issues, cross-border marketing constraints).
- Unsupported transfer or liquidity expectations relative to the actual process and evidence.
- Operational load spikes (investor support, reconciliation exceptions, reporting deadlines).

B.6 Phase 5 — Post-issuance operations & monitoring

Case-specific focus for this phase: Ongoing governance and evidence trails must remain defensible under Luxembourg. Post-issuance compliance must continue to support holders across: Germany, Netherlands. Ongoing operations must reflect the regulated custodian design once its precise role has been independently confirmed. Post-launch monitoring must reflect the actual validated transfer process currently described as ATS or MTF.

Objectives

- Run ongoing compliance, reporting, and investor communications through the instrument's lifecycle.
- Operate lifecycle events (distributions, redemptions, corporate actions) with strong controls and auditability.
- Monitor provider performance, costs, and risk posture; adjust as regulations and market conditions evolve.

Key decisions

- Governance cadence (board/committee oversight, change approvals, vendor reviews).
- Reporting scope and frequency (investor reporting, regulatory reporting, audit readiness).
- Contingency planning (provider exits, custody changes, venue strategy shifts).

Deliverables (checklist)

- Ongoing reporting calendar and control framework (incl. audits and reconciliations).
- Lifecycle event runbooks and evidence retention procedures.
- Vendor performance KPIs and incident log; periodic risk reviews.
- Contingency and change-management playbook (custody/venue migrations, rule updates).

Primary stakeholders

- Issuer operations, compliance, finance; custodian; admin/TA (where applicable).
- Transfer and settlement providers explicitly retained in scope, if any.
- External auditors/advisers for periodic reviews and regulatory updates.

Dependencies / gates

- Stable operating model and evidence trail sufficient for audit/regulatory scrutiny.
- Clear change-management process for token rules, disclosures, and provider configuration.
- Budgeting and resourcing for ongoing operations (often underestimated in early planning).

Risk hotspots

- Compliance drift as rules change or distribution expands across jurisdictions.
- Operational fragility under low-liquidity or stress scenarios (settlement failures, disputes).
- Provider concentration risk and cost creep relative to value delivered.

Appendix C — TVAC Provider Directory (illustrative)

This appendix is an illustrative reference layer intended to help the reader think about which types of external partners may be relevant in a tokenization project.

It is not a recommendation list, procurement shortlist, or ranking. Inclusion or omission does not imply endorsement, criticism, quality judgment, regulatory approval, or case suitability. Any provider mentioned here must be assessed independently for legal fit, jurisdictional coverage, operating model fit, technical capability, conflicts, cost, and execution reliability. This appendix is non-exhaustive and subject to change.

B1. How to use this appendix

Use this appendix to identify the provider categories a case may require, not as a signal that a specific provider should be engaged.

In many cases, the main value of the directory is not the names themselves, but the discipline of asking:

- Which functions must be covered?
- Which functions can be combined?
- Which functions must remain regulated or jurisdiction-specific?
- Which dependencies create concentration risk or vendor-capture risk?
- Which parts of the stack are critical before launch, and which can wait?

B2. Typical provider categories

Depending on the case, a tokenization project may require some or all of the following partner categories:

- legal & regulatory advisers
- tax advisers
- issuance / tokenization platform
- smart-contract / token engineering support
- custody / wallet infrastructure
- transfer agent / registry administration
- fund administration or SPV administration
- trading venue / ATS / MTF / exchange partner
- transfer or liquidity-support provider, only where explicitly required and legally permitted
- KYC / AML / onboarding provider
- payment / settlement / banking partner
- audit / assurance / cybersecurity reviewers
- valuation, data, or oracle-related providers
- distribution / placement / investor onboarding partners
- systems integration / workflow automation support

B3. Illustrative provider examples

The examples below are illustrative only and are grouped by function. They are not recommendations, rankings, or evidence of suitability for this case.

B3.1 Legal & regulatory advisers

These advisers help determine whether the proposed structure is legally enforceable, properly classified, and consistent with the intended issuance, distribution, transfer, custody, and disclosure model.

Typical use cases:

- wrapper design
- securities-law classification

- cross-border marketing analysis
- transfer restriction design
- offering documentation
- legal opinions and regulatory scoping

Illustrative examples:

- CMS
- Cooley
- FIN LAW
- MME
- NÄGELE

B3.2 Issuing platforms

These providers typically support issuance workflows, lifecycle management, permissions, investor access rules, and parts of the reporting or compliance stack.

Typical use cases:

- token issuance
- lifecycle events
- permissions / whitelisting
- cap-table or investor-position management
- compliance rule integration
- reporting workflows

Illustrative examples:

- Aktionariat
- Polymesh (Polymath)
- Realiz Group
- Securitize
- Token City
- Tokeny

B3.3 Custody, trading & settlement

These providers support safekeeping, wallet infrastructure, execution, venue access, settlement, and post-trade operations.

Typical use cases:

- institutional custody
- secure wallet infrastructure
- settlement support
- exchange / venue access
- post-trade servicing
- collateral mobility or transfer control support

Illustrative examples:

- ADDX
- Anchorage Digital
- Archax
- BitGo
- Coinbase Institutional

- Copper
- Ondo Finance
- SIX Digital Exchange (SDX)
- Sygnum Bank
- Taurus

B4. Practical selection principles

When evaluating providers, the reader should typically assess:

- jurisdictional fit
- regulatory status where relevant
- experience with the relevant asset class
- operating-model compatibility
- integration burden
- exit options / portability
- fee model
- service resilience
- auditability and evidence quality
- concentration risk

Where possible, provider choices should reinforce the case logic rather than undermine it. A weak provider stack can easily erode the value thesis of an otherwise promising structure.

B5. Important caution

The right provider stack is highly case-dependent.

A provider that is suitable for a closed institutional collateral workflow may be a poor fit for a cross-border fund wrapper. A provider that is strong in digital-asset custody may not be the right choice for transfer administration or legal structuring. A venue that looks attractive commercially may create regulatory or operating-model problems if the case does not actually need that level of transferability.

Appendix D — Glossary & abbreviations

A practical reference for the most common terms and abbreviations used in this report. Spelling note: tokenization uses “z”.

Term	Meaning
Added Value (AV)	TVAC’s net assessment of whether tokenization appears to improve a case overall once both upside and friction are weighed together. Formula: $AV = (NO + CS + RR) - (TC + NR)$.
AML	Anti-Money Laundering controls intended to detect, prevent, and report suspicious financial activity.
Asset owner / Issuer	The originating party, sponsor, manager, or issuing structure responsible for launching or sponsoring the tokenized case being assessed.
Cash leg	The cash side of a transaction or settlement flow, as distinct from the asset or token side.
Conditional Go	A Tokenization Value Verdict indicating that tokenization may add net value, but remains materially dependent on design discipline, implementation choices, and resolution of key gating conditions.
Custody	The safeguarding, control, and recordkeeping of assets or positions on behalf of investors or market participants.
DeFi	Decentralized Finance. A blockchain-based ecosystem of protocols and applications designed to enable financial activity without relying entirely on traditional centralized intermediaries.
Distribution scope	The intended investor reach of an offering or tokenized structure, including geography, investor type, and access conditions.
DvP	Delivery versus Payment. A settlement model in which transfer of the asset and transfer of payment are linked so that one does not occur without the other.
Economic rights	The financial entitlements attached to a token or instrument, such as income participation, coupon, principal repayment, or profit-sharing.
ESG	Environmental, Social, and Governance factors.
Execution risk	The risk that a concept that appears viable in theory fails in practice because implementation, coordination, economics, or governance proves weaker than assumed.
Governance	The decision-making, control, accountability, and escalation framework surrounding a structure, platform, issuer, or operating model.
Investor segmentation	The way the target investor base is defined and limited by investor type, sophistication, jurisdiction, or eligibility criteria.
KYC	Know Your Customer procedures used to verify customer identity and onboarding eligibility.
Liquidity window	A predefined period or mechanism allowing limited secondary trading, transfer, redemption, or exit opportunities.
MiCA	Markets in Crypto-Assets Regulation in the European Union.
MiFID II	Markets in Financial Instruments Directive II in the European Union.
MTF	Multilateral Trading Facility under EU regulatory frameworks.

Term	Meaning
Net Added Value	See Added Value (AV).
New Opportunities (NO)	Case-specific upside that tokenization creates or materially strengthens, such as new investor reach, new operational utility, collateral use, programmability, or improved strategic flexibility.
New Risks (NR)	Additional risks introduced by tokenization, including legal, regulatory, operational, technological, governance, distribution, or market-structure risks.
NO / CS / RR / TC / NR	Shorthand for TVAC's five factors: New Opportunities, Cost Savings, Risk Reduction, Tokenization Costs, and New Risks.
Non-viable	A Tokenization Value Verdict indicating that the current structure does not support a positive tokenization value case under the TVAC methodology and should not proceed without substantial redesign.
Permissioned	A system design in which access, participation, or transfers are restricted to approved participants under defined rules.
Professional investor	An investor category used in many jurisdictions for parties assumed to have greater sophistication, capacity, or experience than retail investors.
Prospectus	A formal disclosure document that may be required for certain public offerings or investor solicitations under applicable law.
Registry	The official or operational record of holdings, ownership, or investor positions.
Rights mapping	The process of making clear what legal, economic, governance, and transfer rights a token actually represents and how those rights are enforceable.
Risk Reduction (RR)	The extent to which tokenization reduces existing risk in the conventional setup, for example through better controls, improved transparency, clearer auditability, or lower reconciliation burden.
Self-custody	A model in which the end holder directly controls the relevant wallet, keys, or token position without relying on a traditional third-party custodian.
Settlement model	The way asset transfer, cash transfer, record updates, and finality are operationally achieved.
SPV	Special Purpose Vehicle. A ring-fenced legal entity often used for issuance, structuring, asset isolation, or liability segregation.
Tokenization Costs (TC)	The additional costs introduced by tokenization, including structuring, legal work, compliance, integration, platform costs, governance overhead, servicing changes, and operating complexity.
Transfer controls	Rules, permissions, and technical or legal mechanisms governing who may transfer, receive, or hold the token.
Transfer restrictions	Specific limitations placed on secondary transfers, often for regulatory, contractual, or investor-eligibility reasons.
TVAC	Tokenization Value-Add Calculator. A structured decision-support methodology for assessing whether tokenization appears to create net added value in a specific case.
Vendor-capture risk	The risk that a structure becomes economically or operationally overly dependent on one platform, provider, or implementation path in a way that shifts value away from the initiating party.

Term	Meaning
Venue fit	The degree to which the chosen issuance, transfer, trading, or liquidity venue matches the asset, investor scope, regulatory posture, and operating model of the case.
Viable	A Tokenization Value Verdict indicating that the TVAC methodology identifies a positive tokenization value case under the described structure, subject to validation and disciplined execution. It is not an investment, credit, yield, valuation, legal, tax, or suitability assessment.
Whitelist / Whitelisting	A mechanism for restricting participation or transferability to pre-approved investors, wallets, counterparties, or participants.

Appendix E — User-supplied case inputs

E.1 Full verbatim case description

Full original free-text case input provided by the user.

We are a privately owned real estate holding company owning three stabilized residential assets in Berlin and Hamburg through German GmbHs. The properties generate steady rental income but require energy upgrades, façade improvements, and selected tenant-area refurbishments to remain competitive and compliant with evolving ESG standards.

We intend to raise EUR 15–20 million to finance renovations, partially refinance bank debt, and strengthen liquidity buffers. Instead of raising additional traditional mezzanine capital, we are evaluating tokenization through a Luxembourg SPV issuing profit-participating notes linked to net rental cash flow after operating costs and senior debt service.

The tokens would represent clear contractual economic rights, including profit participation and principal repayment, but no operational control over the properties. We would target professional and accredited investors in Germany and the Netherlands only, using a whitelist-based onboarding process with full KYC/AML.

We plan a 12-month lock-up followed by structured quarterly liquidity windows on a regulated venue. Custody would be provided by a regulated custodian. Settlement follows T+1/2 standards aligned with existing capital market workflows. Key considerations include cross-border tax treatment, renovation execution risk, valuation stability, and interest-rate sensitivity. The primary rationale for tokenization is improved investor transparency, fractional access for smaller professional tickets, and long-term administrative efficiency.

E.2 Structured input record

This record reproduces the structured inputs supplied for this assessment. For dropdowns and checkboxes, it also shows the alternatives that were available but not selected.

Field	User response	Other available options not selected
Issuance jurisdiction	Luxembourg	Not applicable
Sale jurisdictions (comma-separated)	Germany, Netherlands	Not applicable
Instrument type	Digital wrapper (CSD/TA/registrar)	Sole-record / native tokenized security
Custody model	Regulated custodian	Transfer agent / CSD; Self-custody (wallets)
Secondary-market liquidity	Quarterly windows	None; Ad-hoc OTC; Continuous
Secondary-market venue	ATS / MTF	DEX (permissioned); OTC; CSD / TA-registered
Payment / cash-leg model	T+1/T+2 securities settlement	Not decided yet; Not material at this stage; Traditional bank / fiat payments; Custodian / bank / TA / CSD settlement; Stablecoin / tokenized cash; Deposit token / tokenized deposit / institutional DvP rail; Advanced: T+0 / atomic DvP; Advanced: off-chain booking
Offering disclosure	Exempt / limited-offering route (jurisdiction-dependent)	KID (PRIIPs); Prospectus
Transfer restrictions	Whitelist + EU gating + 12-month lock-up	Not applicable
Tax considerations (optional)	Cross-border withholding considerations	Not applicable
Investor types	Professional; Qualified / accredited (jurisdiction-dependent)	Retail

An option shown as not selected was not necessarily actively considered or rejected by the user; it was simply not selected for this assessment.

Scenario comparison: Alternative options shown here can help identify design variables worth testing in a new assessment or in TVAC Scenario Lab—for example a different custody model, settlement model, participant type or transfer setup.

Appendix F — Disclaimer

Disclaimer

This report is provided for informational and decision-support purposes only. It does not constitute legal, regulatory, tax, accounting, investment, valuation, or financial advice, and it should not be relied upon as a substitute for professional advice tailored to the specific facts and circumstances of the case.

TVAC is a structured assessment tool designed to evaluate whether and when tokenization may create added value relative to conventional alternatives. Its outputs reflect a methodology-based analytical framework operating under uncertainty. They are not guarantees of commercial success, feasibility, investor demand, regulatory approval, liquidity, execution quality, budget outcome, or risk reduction.

Any scores, verdicts, scenarios, stakeholder assessments, suggested actions, implementation observations, and roadmap elements in this report are indicative only and depend on the quality, completeness, and interpretation of the input data, as well as on assumptions about structure, jurisdiction, service-provider readiness, market conditions, execution capability, and timing.

Tokenization outcomes can differ materially from this assessment if key conditions change, including legal structuring, investor scope, jurisdiction count, custody model, distribution pathway, settlement design, governance arrangements, servicing readiness, technology integration, cost allocation, evidence quality, or liquidity support.

A TVAC verdict is a tokenization added-value verdict only. It must not be presented or used as investment approval, credit rating, yield validation, issuer endorsement, regulatory approval, due diligence report, or investor recommendation. TVAC does not validate projected returns, coupon payments, credit quality, valuations, legal enforceability, tax treatment, or investor suitability.

This report does not replace formal legal analysis, regulatory review, tax analysis, technical architecture review, cybersecurity review, accounting assessment, operational diligence, commercial validation, or board-level decision processes. Independent advice should always be obtained before any structuring, issuance, distribution, investment, operational, or implementation decision is made.

Neither TVAC nor Astoval assumes responsibility for actions taken on the basis of this report alone. Users remain solely responsible for all commercial, legal, regulatory, operational, and strategic decisions related to any proposed tokenization initiative.



Tokenization Value-Add Calculator

A PRODUCT BY ASTOVAL

Astoval provides independent, AI-enhanced early-stage decision support for asset tokenization, helping organizations assess whether tokenization creates genuine added value before significant resources are committed to legal structuring, platform selection or implementation.

PRIVACY AND DATA

Case inputs and generated reports are processed for delivery and are not intentionally retained as a persistent customer-case record in the standard self-service flow. OpenAI API data is not used for model training unless the API customer opts in; separate abuse-monitoring retention may apply for up to 30 days.

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